Raúl Beltrán Gómez

Madrid, Spain (Spanish Citizenship) | +34 642 58 15 03 | rb58853@gmail.com
github.com/rb58853 | portfolio.rb58853.net | linkedin.com/in/rb58853

Professional Summary

AI and software engineer with production experience in computer vision (real-time detection and pose estimation architectures, ONNX and TensorRT deployment, industrial vision) and in LLM systems (MCP servers, agents, NLP, REST APIs). Strong background in Python backends, secure integrations and cross-stack development (React, Unity), and hands-on experience in vision-guided robot control.

Experience

Promart Automatizaciones — Madrid, Spain
Computer Vision and AI Engineer — *01/06/2026 – Present*

- Built the real-time vision system of a robotic teat-sealing cell (rotary milking parlor): CNN object detection of udder and teats, cow identification and stall-number OCR; ONNX/TensorRT GPU inference.
- Developed a 13-keypoint pose-estimation pipeline with temporal smoothing across frames and rule-based geometric checks, designed to minimize false positives.
- Owned the model lifecycle (labeling, augmentation, Optuna tuning, ONNX/TensorRT export, mAP and latency benchmarks, S3 versioning, CI/CD) and its Python stack: clean architecture, FastAPI, tests.
- Linked vision output to robot control via robot APIs; built a Python robot simulator for offline tests.

Cogitia — Remote (Barcelona, Spain)
AI Engineer (LLM Agents and MCP) — *14/05/2025 – 20/05/2026*

- Designed and maintained MCP servers for enterprise workflows, exposing tools, resources and prompts to LLM clients, secured with OAuth 2.1 and header-based authentication.
- Developed LLM agents with LangGraph orchestration for business automation workflows.
- Built NLP modules for automation, including data operations, scheduling and internal query handling.

Word Games — Remote (Bogotá, Colombia)
Game Developer — *20/10/2024 – 10/05/2025*

- Led end-to-end programming of Arcane Ascent (deck-building roguelike, Unity/C#), shipped on Steam.
- Integrated native Steam systems (achievements, saves), improving platform readiness and user experience.

Devman — Remote (Madrid, Spain)
NLP and Backend Developer — *10/04/2024 – 14/10/2024*

- Designed an e-commerce assistant with NLP intent handling and product retrieval for faster discovery.
- Built backend services in FastAPI, including API key security, WebSocket chat and authorization layers.

Education

University of Havana — Havana, Cuba
Bachelor of Computer Science — 23/01/2024

- Coursework: Artificial Intelligence and Machine Learning, Information Retrieval, Compilers, Operating Systems, Databases, Distributed Systems, Networks, Algorithms and Data Structures, Graph Theory.

Skills

Languages: Spanish (native), English (B2).
Programming: Python, C#, C/C++, JavaScript/TypeScript, Assembly (MIPS).
AI/ML: Computer Vision, PyTorch, TensorFlow, Optuna, MCP, LangGraph, LangChain, NLP, RAG.
Software: FastAPI, .NET, React, PyQt6, Unity Engine, Docker, Git.
AI-assisted development: Claude Code (advanced), agentic coding workflows, MCP tool integrations.